

Umid Suleymanov

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Summary

Third-year PhD student in Computer Science at Virginia Tech with 4.5 years of industry experience building end-to-end data pipelines, predictive models, and Business Intelligence solutions. Currently a Graduate Research Assistant at Virginia Tech, and most recently a Data Science Intern at Amazon (Foundational Security), where I ran large-scale analysis in SparkSQL and built agentic LLM systems that extract strong, explainable signals from unstructured text. I excel at translating complex, ambiguous data into strategic decisions using inferential statistics (regression, hypothesis testing, A/B testing), machine learning, agentic LLM systems, and visualization.

Education

Virginia Tech, Blacksburg, VA — *PhD in Computer Science* 2024 – 2027 (Expected)

Advisor: Murat Kantarcioglu, *GPA: 4.0/4.0*

Coursework: Advanced Data and Information Engineering, AI for Software Engineering, Information Visualization

Khazar University, Baku, Azerbaijan — *M.S. in Computer Science* 2019 – 2021

Advisor: Amir Rahmani, *GPA: 4.0/4.0* — Coursework: Machine Learning, Distributed Systems, Software Engineering

ADA University, Baku, Azerbaijan — *B.S. in Computer Science* 2013 – 2018

Advisor: Samir Rustamov, *Cum Laude, GPA: 3.72/4.0* — Coursework: Linear Algebra, Artificial Intelligence, Machine Learning, Data Mining and Decision Support

Professional Experience

Graduate Research Assistant, *Virginia Tech* Aug 2024 – May 2026, Aug 2026 – Present

- Lead original research in LLM safety, *CourtGuard-PMPD* (published at EMNLP 2026 Findings), a policy-grounded multi-agent framework for zero-shot defense against sophisticated jailbreak attacks and automated red-teaming.
- Designed agent-based, inference-time reflective defenses (*SemSIEdit*) *SemSIEdit* in Python and PyTorch that reduced semantic privacy leakage by 35% without degrading model utility.
- Published few-shot learning research (*SSPNet*) for Network Intrusion Detection, outperforming state-of-the-art models by 3.5%. (published at MILCOM 2025).
- Audited fairness-privacy trade-offs of fairness-enhancing algorithms, surfacing subpopulation-level privacy effects (published at EuroS&P 2026).

Data Science Intern, *Amazon* (Foundational Security) May 2026 – Aug 2026

- Developed agentic LLM architectures that autonomously extracted strong, explainable signals from large-scale unstructured security text, converting raw data into actionable insights for business stakeholders.
- Ran large-scale data analysis in SparkSQL across enterprise-scale datasets; applied statistical methods and root-cause analysis to troubleshoot data anomalies and propose strategic resolutions.
- Conducted and published applied research on LLM safety and security analytics.

Data Science Intern, *Amazon* (AWS Proactive Security) May 2025 – Aug 2025

- Built robust ETL pipelines, machine learning models, and automated reporting metrics using Python, SQL, and AWS-native tools to deliver actionable insights to business stakeholders.
- Developed BI solutions and dashboards (QuickSight) to track proactive security metrics, optimizing BI processes for speed and reliability.

Instructor of Computer and Information Sciences, *ADA University* Jan 2023 – Aug 2024

- Instructed courses on Big Data Engineering, Deep Learning, and Data & Information Engineering; designed syllabi, instructional materials, and assessments using Blackboard LMS.
- Researched pre-training and efficient fine-tuning of LLMs for low-resource languages and Explainable AI (XAI) at the Center for Data Analytics and Research.

Leading Data Scientist, *E-Gov Development Center* Jun 2022 – Jan 2023

- Led design and deployment of end-to-end data pipelines and BI solutions, translating business requirements into actionable data science strategies.
- Built custom Elasticsearch text analyzers and used A/B testing and statistical methods to measure search relevance, increasing click-through rates by 6%.

Senior Data Scientist, *E-Gov Development Center* Nov 2019 – Jun 2022

- Designed predictive models and analytics reports to optimize queue management, using regression techniques to

reduce related customer complaints by 16%.

- Automated a document anonymization pipeline with custom NER, accelerating processing speed by 32% while ensuring stable, maintainable code.

Data Engineer, E-Gov Development Center

Jul 2018 – Nov 2019

- Designed relational data structures (DDL/schema definition) and implemented robust ETL routines for large-scale data acquisition, transformation, and storage, improving data reliability and quality.

Selected Publications

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- [1] U. Suleymanov, R. Bayramov, S. Gafarli, T. Mammadov, S. Musayeva, M. Kantarcioglu. “CourtGuard-PMPD: Zero-Shot Policy Adaptation in LLM Safety without Fine-Tuning using Prompt-Based Modular Policy Database”, EMNLP, 2026 Findings.
 - [2] U. Suleymanov, Ilhama Noruzova, et al. “Auditing Fairness–Privacy Trade-offs: Subpopulation-Level Effects of Fairness-Enhancing Algorithms”, Accepted at EuroS&P, 2026.
 - [3] A. Asgarov, U. Suleymanov, A. Khatri. “SIGMA: Search-Augmented On-Demand Knowledge Integration for Agentic Mathematical Reasoning.” *AAAI (LMReasoning)*, 2026.
 - [4] U. Suleymanov, et al. “SSPNet: Semi-Supervised Prototypical Networks for Few-Shot Network Intrusion Detection.” *IEEE Military Communications Conference (MILCOM)*, 2025.
 - [5] U. Suleymanov, Murat Kantarcioglu, et al. “SPRINT: Semi-supervised Prototypical Representation for Few-Shot Class-Incremental Tabular Learning”, Under Review, 2026.
 - [6] U. Suleymanov, et al. “Beyond Refusal: Probing the Limits of Agentic Self-Correction for Semantic Sensitive Information”, Under Review, 2026.

Invited Talks & Academic Service

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- **Reviewer** at NeurIPS, ACM Transactions on Privacy and Security, and AICT; **Gold Reviewer** at ICML 2026.
 - **Speaker**, “Large Language Models: Reasoning and Security”, Data, ERP and AI Summit. Oct 2025
 - **Judge (CS Domain)**, Intel International Science and Engineering Fair (ISEF) Local Selection Feb 2023
 - **Tutorial**, “Intermediate Machine Learning,” Kaggle 30 Days of ML, Google Developers Group Baku Aug 2021

Technical Skills

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- **Programming:** Python, SQL, SparkSQL, R, Java
 - **ML/Stats:** Inferential Statistics, Regression, Hypothesis Testing, A/B Testing, Few-Shot Learning, NLP, Supervised/Unsupervised Learning
 - **Deep Learning:** PyTorch, TensorFlow, Keras, HuggingFace Transformers, vLLM
 - **AI Safety & Security:** LLM Red-Teaming, Jailbreak Defenses, Anomaly Detection, Privacy Auditing
 - **Infrastructure & Data:** Linux, Docker, Git, AWS S3/SageMaker, Spark, BigQuery, Relational Databases

Honors & Achievements

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- Finalist, International Data Analysis Olympiad
 - TensorFlow Developer Certificate
 - First Place, AzInTelecom Hackathon
 - Instructor, 30 Days of ML (Kaggle/Google, 2021)
 - 8+ peer-reviewed papers in AI/ML
 - Honorable Mention, ACM ICPC Contest